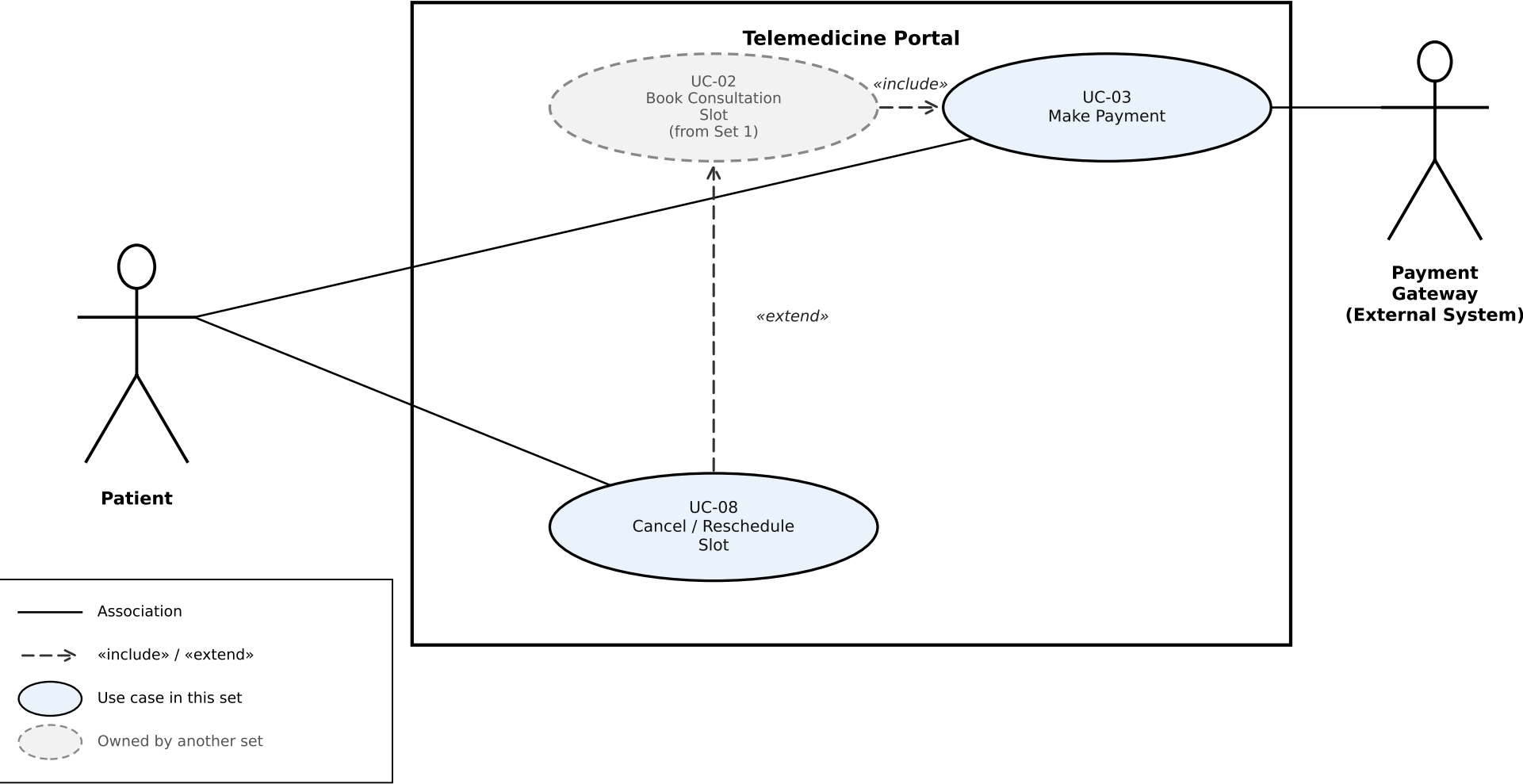


Set 2 — Payment & Booking Changes

Weightage: 25% | Use cases: UC-03, UC-08



Use-Case Flow Specification

Telemedicine Slot Booking & Prescription Portal

Set 2 — Payment & Booking Changes | Weightage: 25%

Use Case: UC-08 — Cancel / Reschedule Slot (*extends UC-02: Book Consultation Slot, owned by Set 1*)

Primary Actor

Patient

Preconditions

- Patient is authenticated and has at least one confirmed, upcoming booking.
- The consultation has not yet started.
- The cancellation/reschedule window defined by clinic policy is still open.

Postconditions

- Success (cancel): The slot is released back to the pool, the physician is notified, and any refund due is initiated.
- Success (reschedule): The original slot is released, a new slot is reserved against the same payment, and both parties are notified.
- Failure: The original booking remains unchanged and the patient is told why the request was rejected.

Main Success Scenario

1. Patient opens “My Appointments” and selects a confirmed upcoming booking.
2. System displays the booking details along with the applicable cancellation and refund policy.
3. Patient chooses either “Cancel” or “Reschedule”.
4. System validates the request against the clinic's cut-off window.
5. If rescheduling, the patient picks a new available slot for the same physician and the system reserves it against the existing payment.
6. If cancelling, the system releases the slot and initiates a refund through the Payment Gateway per the policy tier.
7. System notifies the attending physician of the change.
8. System sends the patient a confirmation with the updated booking reference or refund reference.
9. Use case ends successfully.

Alternate Flow — 4a. Request Outside the Cut-Off Window

- a. 4a1. System detects the request falls inside the no-change window (for example, under 2 hours before the appointment).
- b. 4a2. System rejects the change and displays the policy reason along with the non-refundable amount.
- c. 4a3. Patient may either keep the original booking or cancel without refund; the booking record is updated accordingly and the use case ends.

Requirements Table

Telemedicine Slot Booking & Prescription Portal

Actors: Patient, Attending Physician, Payment Gateway (external system).

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale	Set
FR-001	Functional	The system shall enable physicians to author and digitally sign structured prescriptions that are automatically compiled into downloadable PDF records.	High	Pass: Valid cryptographic signature attached to prescription PDF. Fail: Prescription generated without doctor license validation.	Ensures prescriptions are legally valid, tamper-evident, and traceable to a licensed physician.	Set 4
FR-002	Functional	The system shall allow a patient to search and filter available physicians by medical specialty and real-time slot availability.	High	Pass: Results show only physicians with ≥ 1 open slot in the next 7 days, correctly filtered by specialty. Fail: Results include unavailable or wrong-specialty physicians.	Core discovery function patients need before booking; reduces failed booking attempts.	Set 1
FR-003	Functional	The system shall allow a patient to book a specific video consultation slot, reserving it exclusively for that patient once payment is confirmed.	High	Pass: The booked slot disappears from all other patients' availability lists within 2 seconds. Fail: The same slot can be booked by two different patients.	Prevents double-booking and guarantees the integrity of the appointment schedule.	Sets 1, 2
FR-004	Functional	The system shall automatically generate and transmit a unique, encrypted video-consultation room link to both patient and physician no later than 10 minutes before the scheduled appointment.	High	Pass: Both parties receive a working, single-use encrypted link ≥ 10 minutes before start time. Fail: Link is missing, reused across sessions, or delivered late.	Ensures secure, timely, and private access to the tele-consultation room.	Set 3
FR-005	Functional	The system shall allow a patient to download their digitally signed prescription as a PDF from their account within 24 hours of consultation completion.	Medium	Pass: Patient downloads a valid, signed PDF from the dashboard within 24 hours of consultation end. Fail: PDF is unavailable, unsigned, or late.	Gives patients timely access to prescriptions required for pharmacy fulfillment.	Set 4
NFR-001	Nonfunctional (Performance & Security)	All video-session metadata and patient clinical notes must be encrypted in transit using TLS 1.3 and at rest using AES-256.	High	Pass: Benchmarking confirms target latency and security standards under simulated peak load. Fail: Any data found unencrypted in transit or at rest.	Protects sensitive patient health data (PHI) and satisfies healthcare data-protection obligations.	All sets
NFR-002	Nonfunctional (Performance & Scalability)	The system shall support at least 500 concurrent video-consultation sessions with end-to-end video latency under 300 ms.	High	Pass: Load test at 500 concurrent sessions shows average latency < 300 ms with zero session drops. Fail: Latency exceeds 300 ms or sessions fail under peak load.	Guarantees usable, real-time video quality during peak clinic hours.	Set 3
NFR-003	Nonfunctional (Availability & Reliability)	The system shall maintain at least 99.9% uptime during published clinic hours, with no confirmed booking or signed prescription lost during a failover.	High	Pass: 30-day monitoring shows ≤ 43 min downtime and a failover drill loses zero records. Fail: Uptime below 99.9% or any record lost.	Missed consultations and lost prescriptions carry direct clinical risk.	All sets

Module Split & Weightage

The eight use cases are divided into four equally weighted sets. Each set has its own use-case diagram and its own one-page use-case flow specification.

Set	Module Name	Use Cases Owned	Relationship Modelled	Weightage
Set 1	Doctor Discovery & Slot Booking	UC-01 Search Doctor by Specialty; UC-02 Book Consultation Slot	UC-02 «include» UC-03	25%
Set 2	Payment & Booking Changes	UC-03 Make Payment; UC-08 Cancel / Reschedule Slot	UC-02 «include» UC-03; UC-08 «extend» UC-02	25%
Set 3	Tele-Consultation Session	UC-04 Generate Encrypted Video Link; UC-05 Conduct Video Consultation	UC-05 «include» UC-04	25%
Set 4	Prescription Lifecycle	UC-06 Author & Digitally Sign Prescription; UC-07 Download Prescription PDF	UC-06 «extend» UC-05	25%